How to produce good seed

Ten steps for farmers to produce their own good seed

1. Select a fertile field.
2. Use clean, good quality seed.
3. Plow, puddle and level the field well to control weeds and improve water management.
4. If transplanting, plant young (15−20 d) seedlings from a healthy, weed-free nursery at two per hill at 22.5 cm x 22.5 cm spacing.
5. Apply balanced nutrients (Nitrogen, Phosphorous, Potassium, Sulfur, and Zinc) as per crop demand.
6. Keep the crop free of weeds, insect pests and diseases.
7. At maximum tillering and flowering, rogue off-types (by plant height, appearance, flowering time, etc.) and poor, diseased or insect damaged plants, or plants with discolored panicles.
8. Harvest at full maturity and 20−25% moisture content (80−85% of the grains are straw-colored).
9. Thresh, clean, dry (12−14% moisture content), grade and label the harvested seed.
10. Store the labeled seed in sealed clean containers placed in a cool, dry, and clean area.

Winnowing to get good seed

Harvested seed includes seed of varying sizes and non-seed matter (e.g., weeds and trash). Full plump (heavier) seed can be selected by winnowing with natural wind or an electric fan.

Procedure: Pour seed slowly from a height of 1−1.5 m.

Repeat winnowing, if necessary. Select heavier seed closer to the side from which the wind blows. This procedure will also remove lighter weed seed and non-seed matter.

Winnowing removes lighter seeds and debris.

Drying and storing good seed

After harvest, clean seed and select full and uniform seed.

Dry seed to 12−14% moisture content. Store the seed in sealed airtight containers until ready for planting (seed is good for up to one year if stored properly). Seed in non-airtight containers absorbs moisture and loses viability over time.

Read: Postharvest - Drying | Storage

How to prepare the seedlings for transplanting

Prior to transplanting, seedlings need to be raised in a nursery. Seedling nurseries usually use 5−10% of the total farming area.

When choosing the appropriate nursing system, consider the availability of sunlight, water, labor, land, and agricultural implements.

Wet-bed

Use this method in areas with sufficient water supply. Allot 1/10 of the field for the seed bed area, and prepare 40 kg of seed to transplant 1 ha of land.

1. Prepare beds at 1 m wide by convenient length. Raise the soil to 5−10 cm height.
2. Broadcast pre-germinated seeds in thoroughly puddled and leveled soil.
3. Construct drainage canals for proper water removal.
4. Add organic manure (decompose) and a small amount of inorganic fertilizer as basal dressing. This increases seed vigor and allows easier uprooting for transplanting.
5. Transplant seedlings at 15−21 days old.

Dry-bed

Prepare the nursery in dry soil conditions. Ensure that the site is free of shade and has access to irrigation facilities. Allot 1/10 of the field for the seed bed area, and prepare 60−80 kg of seed to transplant 1 ha of land.

1. Prepare beds at 1 m wide by convenient length. Raise the soil to 5−10 cm in height.
2. Distribute a layer of half burned paddy husk on the nursery bed to facilitate uprooting.
3. Prevent moisture stress by irrigation. Without appropriate moisture, roots may be damaged during pulling.
4. If nutrient supply is low, apply basal fertilizer mixture and incorporate it between rows.
5. Transplant seedlings at 15−21 days old.

Seedlings raised in dry-bed are short, strong, and has a longer root system than those raised in wet-beds.

Dapog

Dapog or mat method is most appropriate for growing short duration varieties, as seedlings experience less transplanting shock.

Compared to other methods, this requires less labor, and has minimal root damage.

Prepare dapog nurseries where a flat firm surface is available and water supply is very reliable. Allot 100 m^2/ha or 1% of the field for the seedbed, and prepare 40−50 kg of seed per ha.

1. Mark out 1 m wide and 10−20 m long plots.

2. Cover the surface with banana leaves, plastic sheets, or any flexible material from penetrating the bottom layer of the soil. Cemented floors may also be used as base. Form the boundary with bamboo splits or banana sheath.
3. Cover the seedbed with about 1 cm of burned paddy husk or compost.
4. Sow pre-germinated seeds on the seedbed. Maintain a thickness of 5−6 seeds (1 kg per 1.5 m^2).
5. Sprinkle water to the seeds after sowing, and then press down by hand or with a wooden flat board.
6. Prevent water stress by irrigation.
7. Transplant seedlings at 9−14 days old.

Modified mat nursery

The modified mat nursery uses less land and requires fewer seeds and inputs (i.e., fertilizer and water). Allot 100 m^2/ha for the seedbed, and prepare 18−25 kg of good quality seeds.

1. Cover the surface of 4 cm layer soil mix with banana leaves, plastic sheets, or any flexible material from penetrating the bottom layer of the soil.
2. Sow pre-germinated seeds on the seedbed, then sprinkle with water. Maintain a thickness of 2−3 seeds.
3. Water the nursery 2 times a day for 5 days.
4. Transplant seedlings at 15−21 days old, when seedlings reach the four-leaf stage.

Bubble tray nursery

The bubble tray nursery follows the modified mat system to develop seedlings with root balls. Seedlings raised on bubble trays experience less transplanting shock. At 12−15 days old, seedlings are broadcast in the field.

To do this:

1. Prepare 750 trays per hectare of paddy.
2. Raise seedlings on plastic trays of 59 cm x 34 cm with 434 embedded holes.

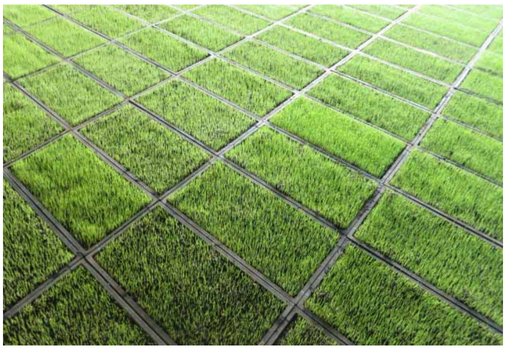
Fact sheet: How to prepare the modified mat nursery

Read: How to control weeds in the nursery

How to prepare seedlings for mechanical transplanting

Mechanical transplanters have built-in trays or seedling boxes.

Grow seedlings on a thin layer of soil in 30 cm x 60 cm trays per seedling box. In some instances, seedlings are grown on larger areas and then cut into rectangular strips (mats of seedlings) that fit into the planting trays of the transplanter.



How to manage soil fertility

Print this page

Applying nutrients to the crop is essential in managing soil fertility so the plants grow and develop normally. A number of crop problems can be related to inefficient management of nutrients and nutrient imbalances in the field.

Site-Specific Nutrient Management

Site-specific nutrient management (SSNM) enables farmers to dynamically adjust fertilizer use, by supplying optimum amounts of nutrients at critical time points in the crop's growth to produce high yields.

In SSNM, farmers tailor their nutrient management strategy to the specific conditions of their field.

The following are steps in SSNM:

STEP 1 Establish an attainable yield target

Identify estimated yield based on location and season. Consider factors such as climate, rice cultivar, and crop management.

The yield target determines the total amount of nutrients that must be taken up by the crop.

STEP 2 Effectively use existing nutrients

Indigenous nutrients which comes from the soil, along with organic materials, crop residues, manures and irrigation, need to be managed properly to achieve optimal crop nutrient uptake.

Read: Use crop residues as mulches | How to make a compost | What are cover crops

STEP 3 Apply fertilizer to fill in other nutritional needs of the crop

NPK fertilizers are applied to supplement indigenous nutrients.

The quantity of application is determined by the target yield and the amount of nutrients needed by the crop.

To assess the crop's nutrient needs, use the Leaf Color Chart.

Fact sheets: Nitrogen (N) | Phosphorus (P) | Potassium (K) | Zinc (Zn)

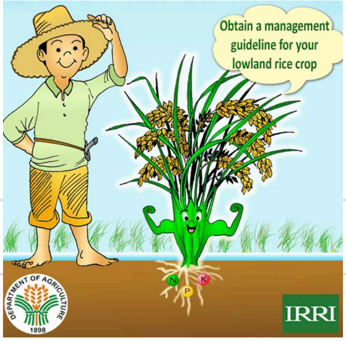
Go to web app: Rice Crop Manager

Read: for researchers and scientists - SSNM explained | SSNM in detail

Fact sheets: for farmer-managed research - Addition plots | Nitrogen split applications | Nutrient omission plots

Crop Manager

Rice Crop Manager

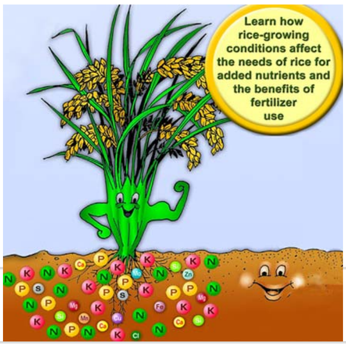


Rice Crop Manager (RCM) is a computer- and mobile phone-based application that provides extension agents and farmers with advice on crop and nutrient management matching their particular farming conditions.

Location-specific guidelines are currently available in Bangladesh, China, India, Indonesia, the Philippines, and West Africa.

Go to web app: Rice Crop Manager | RCM Tutorial

Nutrient Teacher for Rice



Nutrient Teacher for Rice is a teaching tool on RCM. This app is made for students and instructors of introductory courses in soil science and crop science. It can also be used by researchers.

It shows how information on season, crop establishment, variety, growth duration of rice, yield, residue management, soil fertility, and use of organic materials as sources of nutrients affect rates of nitrogen (N), phosphorus (P), and potassium (K) fertilizer.

Go to web app: Nutrient Teacher for Rice

Did this page help you?

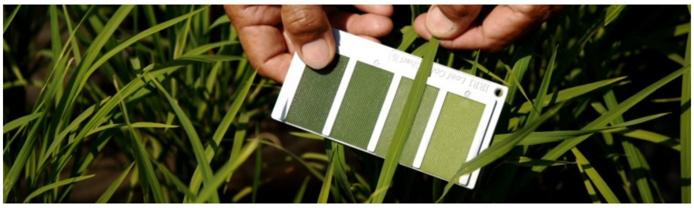
Cover crops

Growing cover crops help suppress weeds and enrich the soil.
Read more »

Deficiencies and toxicities

Nutrients are important in rice crop growth and development. Having too much or too less of the nutrients can affect the yield.
Read more »

Leaf Color Chart



Plants need optimum amounts of fertilizers in order to develop properly. Use the Leaf Color Chart (LCC) to estimate the rice crop's nitrogen needs.
Read more »

Storage

Print this page

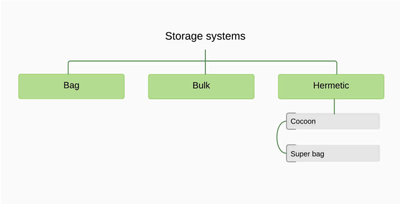
The purpose of any grain storage facility is to provide safe storage conditions for the grain in order to prevent grain loss caused by adverse weather, moisture, rodents, birds, insects and micro-organisms like fungi.

In general, it is recommended that rice for food purposes be stored in paddy form rather than milled rice as the husk provides some protection against insects and helps prevent quality deterioration.

However, when rice can be stored as brown rice, 20% less storage capacity will be needed. **Brown rice** is rice grain with its hulls removed but not polished. Under tropical conditions brown rice has a very short shelf life, approximately two weeks.

Storage systems

Rice storage facilities take many forms depending on the quantity of grain to be stored, the purpose of storage, and the location of the store.



Storage systems can be through bag, bulk, or hermetic containers.

- Bag storage- grain is stored in 40−80 kg bags made from either jute or woven plastic
- Bulk storage - grain is stored in bulk at the farm or at commercial collection houses
- Hermetic storage - grain is stored in an airtight container so that that moisture content of the stored grain will remain the same as when it was sealed. These storages can extend germination life of seeds, control insect grain pests, and improve headrice recovery. Examples include:
  - IRRI Superbag - available to farmers and processors at low cost
  - Cocoon - commercially available
  - Other locally available containers - useful in rural settings, where local containers can be easily converted into hermetic storage systems

Read more »

Guidelines for safe storage

Good storage systems include (1) protection from insects, rodents and birds, (2) ease of loading and unloading, (3) efficient use of space, (4) ease of maintenance and management, and (5) prevention of moisture re-entering the grain after drying.

Safe storage of rice for longer periods is possible if three conditions are met:

1. Grain is maintained at moisture levels of 14% or less and seed is stored at 12% or less
2. Grain is protected from insects, rodents and birds
3. Grain is protected from re-wetting by rain or imbibing moisture from the surrounding air

The longer the grain needs to be stored, the lower the required moisture content will need to be. Grain and seed stored at moisture contents above 14% may experience the growth of molds, rapid loss of viability and a reduction in eating quality.

Good hygiene in the grain store or storage depot is important in maintaining grain and seed quality. To maintain good hygiene in storage:

- Keep storage areas clean. This means sweeping the floor, removing cobwebs and dust, and collecting and removing any grain spills.
- Clean storage rooms after they are emptied and this may include spraying walls, crevices and wooden pallets with an insecticide before using them again.
- Placing rat-traps and barriers in drying and storage areas. Cats deter and help control rats and mice.
- Inspect storage room regularly to keep it vermin proof.
- Inspect the stored seeds once a week for signs of insect infestation. When necessary and only under the direction of a trained pest control technician, the storage room or the seed stock may be sealed with tarpaulin and treated with fumigants.

Specific challenges in the humid tropics

Rice grain is hygroscopic and in open storage systems the grain moisture content will eventually equilibrate with the surrounding air at the so called equilibrium moisture content (EMC). High relative humidity and high temperatures typical for the humid tropical climate lead to grains absorbing water in storage and to a high final moisture content.

> In many tropical countries, the equilibrium moisture content is above safe storage moisture levels.

Did this page help you?

Storage systems

Rice can be stored in bags, bulk, or hermetic containers.
Read more »

Storage pests

Storage pests cause losses through a combination of feeding, spoiling and contamination of both paddy and milled grain.
Read more »

Cocoon

The Cocoon™ is a commercially available hermetic storage container that consists of two plastic halfs that are joined together with an air-tight zipper.
Read more »

Measuring moisture content

Different moisture contents are required per postproduction operation. These should be followed to ensure good quality of paddy.
Read more »

IRRI Super Bag

Using local containers

The IRRI Super bag makes the principle of hermetic storage available to farmers and processors at low cost.
Read more »

In local markets there are available containers that can be easily converted into a hermetic storage system.
Read more »

FAQs on storage

Answers to frequently asked questions on storage
Read more »

How to manage water

Print this page

Rice is typically grown in bunded fields that are continuously flooded up to 7−10 days before harvest.

Continuous flooding helps ensure sufficient water and control weeds.

Lowland rice requires a lot of water.

On average, it takes 1,432 liters of water to produce 1 kg of rice in an irrigated lowland production system. Total seasonal water input to rice fields varies from as little as 400 mm in heavy clay soils with shallow groundwater tables to more than 2000 mm in coarse-textured (sandy or loamy) soils with deep groundwater tables.

> Around 1300−1500 mm is a typical amount of water needed for irrigated rice in Asia. Irrigated rice receives an estimated 34−43% of the total world's irrigation water, or about 24−30% of the entire world's developed fresh water resources.

Worldwide, water for agriculture is becoming increasingly scarce. Due to its semi-aquatic ancestry, rice is extremely sensitive to water shortages.

To effectively and efficiently use water and maximize rice yields, the following good water management practices can be done:

STEP 1 Construct field channels to control the flow of water to and from your field

The construction of separate channels to move water to and from each field greatly improves the control of water by individual farmers.

Field channels allow water to be delivered to the individual seed beds separately and the main field does not need to be irrigated until it's time to plant in the main field.

In addition, the ability to control water to your field is important when you need to retain water (especially after applying fertilizer so nutrients are not lost) or when you need to drain the field for harvest.

Construction of individual field channels is the recommended practice in any type of irrigation system.

STEP 2 Prepare the land to minimize water loss and create a hard pan

Large amounts of water can be lost during land soaking prior to puddling when large and deep cracks are present due to drainage of water down the cracks, beyond the root zone.

Till the soil to fill cracks

Perform shallow tillage operations before land soaking. This fills in the cracks and can greatly reduce the amount of water used in land preparation.

Puddle the field to reduce water loss

- For clayey soils that form cracks during the fallow period, puddling results in a good compacted hard pan
- For coarse sandy soils, puddling may not be effective

- For heavy clay soils, puddling may not be necessary to reduce water losses because of the low infiltration rate of such soils; however, puddling may still be necessary if the soil was cracked prior to primary tillage, if weeds are present prior to transplanting, or if the soil is too hard or cloddy for transplanting after soaking

Despite reducing water loss, the action of puddling itself consumes water. There is a trade-off between the amount of water used for puddling and the amount of water “saved” during the crop growth period because of a compact hard pan.

Wet land preparation can consume up to a third of the total water required for growing rice in an irrigated production system.

If water cost or availability at the time of crop establishment is a concern, consider dry land preparation which uses considerably less water than wet land preparation.

Minimize time between operations to reduce water use

In some canal irrigation systems, the period of time between land soaking for land preparation and planting can be up to 40 days. To minimize time between operations:

- install field channels
- use common/community seed beds
- plant nearby fields at the same time, or
- practice direct wet seeding

STEP 3 Level the field

A well-leveled field is crucial to good water management. An unleveled field requires an extra 80−100 mm of water to give complete water coverage. This is nearly an extra 10% of the total water requirement to grow the crop.

Most fields need to be plowed twice before you can level. In wet land preparation, the second plowing should be done with standing water in the field to define high and low areas.

Read: Leveling implements | Laser leveling

STEP 4 Construct bunds and repair any cracks or holes

Good bunds are a prerequisite to limit water losses. Bunds should be well compacted and cracks or rat holes should be plastered with mud at the beginning of the crop season to limit water loss.

Bunds should be high enough (at least 20 cm) to avoid overflowing during heavy rainfall.

Lower levees of 5−10 cm height in the bunds can be used to keep the ponded water depth at that height. These levees can be heightened with soil when more stored water is needed.

Read: How to construct bunds

Different crop establishment methods require different water management practices:

For continuous flooding

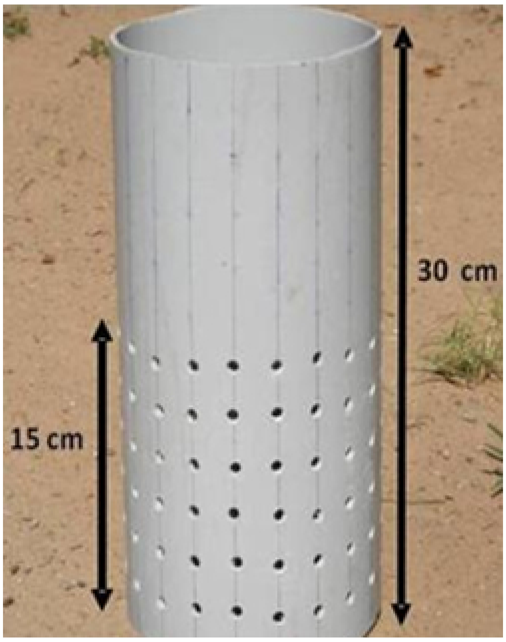
Continuous flooding of water generally provides the best growth environment for rice.

After transplanting, water levels should be around 3 cm initially, and gradually increase to 5−10 cm (with increasing plant height) and remain there until the field is drained 7−10 days before harvest.

For direct wet seeded rice, field should be flooded only once the plants are large enough to withstand shallow flooding (3-4 leaf stage).

For safe Alternate Wetting and Drying

Transplanting



Alternate Wetting and Drying (AWD) can be started a few weeks (1-2) after transplanting. Irrigate and then allow the water depth to drop to 15 cm below the surface using a field water tube (pictured to the right) to monitor the water level depth. Once the water level has dropped to 15 cm below the surface, re-flood the field to a depth of 5 cm above the surface and repeat. From one week before to one week after flowering, the field should remain flooded. After flowering, during grain filling and ripening, the water level can drop to 15 cm below the surface before re-flooding.

When many weeds are present, AWD should be post-poned for 2-3 weeks to assist suppression of weeds by ponded water and to improve the efficacy of herbicide.

Policy brief: AWD reduces green house gas emissions and saves water

Direct seeded rice

Keep the soil moist but not saturated from sowing till emergence, to avoid seeds from rotting in the soil.

After sowing, apply a flush irrigation to wet the soil, if there is no rainfall.

Saturate the soil when plants have developed three leaves, and then follow the safe Alternate Wetting and Drying practices as described above.

Read: Direct seeding

Lowland rice is extremely sensitive to water shortage (below saturation) at the flowering stage. Drought at flowering results in yield loss from increased spikelet sterility, thus fewer grains.

Keep the water level in the fields at 5 cm at all times from heading to the end of flowering.

In case of water scarcity, apply water-saving technologies such as Alternate Wetting and Drying (AWD) and consider changing planting method from puddled transplanting to non-puddled transplanting or dry-direct seeding.

Fact sheets: Aerobic Rice | Alternate Wetting and Drying (AWD)
FAQ: What's the difference between dry seeded rice (DSR) and aerobic rice?

Did this page help you?